

Building Program Calculations

Project: Commercial Building

Gross Square Feet (GSF) = 100,000 SF

4 Levels (3 Levels above grade + U/G Parking)

$$\begin{aligned}\text{Area per Floor (Footprint)} &= \text{Gross Area} / \# \text{ Levels} = 100,000 \text{ SF} / 4 = \\ &= 25,000 \text{ SF/Level}\end{aligned}$$

Structural Bays Size

$$\begin{aligned}\text{Max 8 m span} \\ 8 \text{ m} \times 3.28 \text{ ft/m} &= 26.2 \text{ ft (max)}\end{aligned}$$

Parking

$$\begin{aligned}\text{Parking aisle } &18 \text{ ft} - 22 \text{ ft} \\ \text{Pricing database } &20 \text{ ft or } 25 \text{ ft} \\ \text{Use } &20 \text{ ft bays}\end{aligned}$$

Parking requires 5 bays wide

$$\begin{aligned}5 \text{ bays} \times 20 \text{ ft/bay} &= 100 \text{ ft wide} \\ 25,000 \text{ SF} / 100 \text{ ft} &= 250 \text{ ft} \\ \text{Approximate building dimensions: } &250 \text{ ft} \times 100 \text{ ft}\end{aligned}$$

Building length

$$\begin{aligned}25,000 \text{ SF} / 100 \text{ ft} &= 250 \text{ ft} / 20 \text{ ft/bay} = 12.5 \text{ bays} \Rightarrow 12 \text{ bays} \\ 12 \text{ bays} \times 20 \text{ ft/bay} &= 240 \text{ ft length} \\ \text{GSF} &= 100 \text{ ft} \times 240 \text{ ft} \times 4 \text{ levels} = 96,000 \text{ SF} \\ \text{Area per Floor (Footprint)} &= \text{Gross Area} / \# \text{ Levels} = 96,000 \text{ SF} / 4 = \\ &= 24,000 \text{ SF/level} \\ \text{Useable space ratio} &= (1.3 + 1.3 + 1.2 + 1.2) / 4 = 1.25 \\ \text{Useable space} &= \text{Gross Area} / \text{US ratio} \\ 24,000 \text{ SF/level} / 1.25 &= 19,200 \text{ USF/Level} \\ \text{Core Area} &= 24,000 \text{ SF/Level} - 19,200 \text{ USF/Level} = 4,800 \text{ SF/Level} \\ \text{Length} &= \text{Core Area} / \text{Width}\end{aligned}$$

4,800 SF/Level / 20 ft = 240 ft = total length (240 ft not good)

4,800 SF/Level / 30 ft = 160 ft

Core Area:

160 ft / 20 ft/bay = 8 bays length

30 ft / 20 ft/bay = 1.5 bays wide

Tenant place- Ceiling height

Cafeteria - 10'

Childcare Center - 9'

Conference Rooms - 12'

Health Clinics - 9'

Offices - Fully Open - 9'

Physical Fitness - 12'

Retail Outlets - 12'

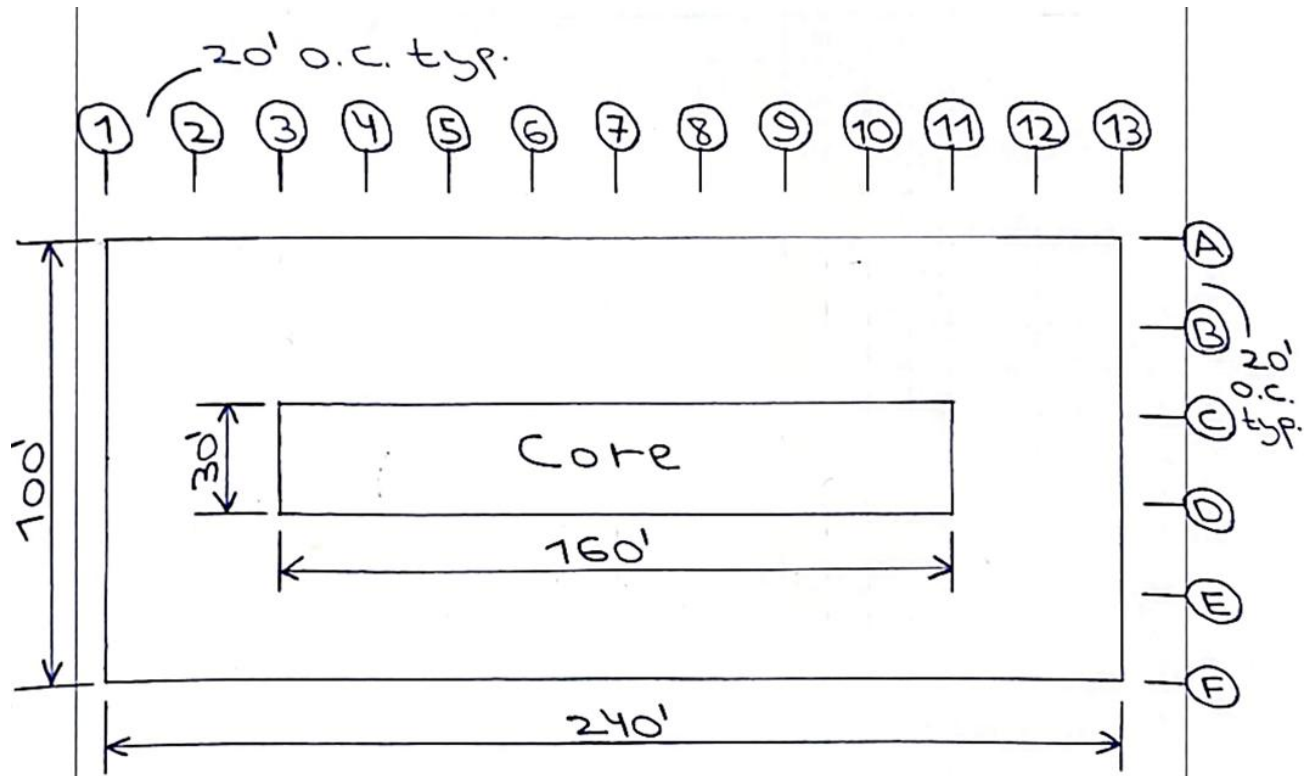
Table 1 - Building Program Summary

Floor	Use 1	SF	Use 2	SF	Tenant Space (SF)	Common Core (SF)	Total Gross (SF)
Level 3	Offices	15,000	Conference Rooms	4,200	19,200	4,800	24,000
Level 2	Health Clinic	11,000	Physical Fitness Center	8,200	19,200	4,800	24,000
Level 1	Retail	13,500	Cafeteria	5,700	19,200	4,800	24,000
U/G	Parking	19,200			19,200	4,800	24,000
				Totals	76,800	19,200	96,000

Table 2 - Building Elevation Preliminary Calculations

Description	Δ Elevation	Actual Elevation
Roof		46.2'
Structure	+1.7'	+46.2'
MEP	+1.7'	+44.5'
Ceiling Max	+12'	+42.8'
Level 3		30.8'
Structure	+1.7'	+30.8'
MEP	+1.7'	+29.1'
Ceiling Max	+12'	+27.4'
Level 2		15.4'
Structure	+1.7'	+15.4'
MEP	+1.7'	+13.7'
Ceiling Max	+12'	+12'
Level 1		0.0'
Structure	-1'	-1'
MEP	-1.75'	-2.75'
Clearance	-7'	-9.75'
Level 0 (U/G)		-9.75'

Proposed Floor + Grid Layout



Gross Area Building = 100 ft x 240 ft = 24,000 SF

Area Core = 30 ft x 160 ft = 4,800 SF